

Stable Counterfactual Identity Carries an Information Rate

Jan Mikulík
(Dated: August 2026)

An exact stochastic interface does not determine a unique latent realization, and different realization contracts need not have the same information cost. We compare two finite-horizon exact contracts for a two-action memoryless channel. In a history-indexed causal response tree, counterfactual events may depend on the realized past; in a synchronous stable-event table, the same time-action label must denote the same latent event across all histories. We show that the first contract reaches the Shannon floor, $C_\infty^{\text{causal}}(P, Q) = \max\{H(P), H(Q)\}$, whereas the second obeys an information-spectrum lower bound. Their difference therefore satisfies

$$\tau_{\text{id}}(P, Q) \geq \frac{1}{2}[W_1(\mu_P, \mu_Q) - |H(P) - H(Q)|],$$

where μ_P, μ_Q are the one-symbol surprisal laws. Equal-entropy actions with different probability spectra consequently pay a strictly positive asymptotic rate for stable cross-history event identity. The result prices a realization contract; it does not identify a physical substrate.

A stochastic law specifies what may be observed. It does not, by itself, specify which mutually exclusive possibilities must be represented by one common latent object. This distinction appears in exact channel synthesis, common-information problems, causal transport, and random-map realizations of stochastic dynamics [4–6]. Here we isolate a particularly sharp instance: two latent constructions reproduce *every adaptive observable policy exactly*, yet their minimum asymptotic seed entropies differ by a strictly positive rate. We make no historical priority claim for the individual ingredients—minimum-entropy coupling, information-spectrum converses, or causal transport. The contribution of this Letter is the explicit contract comparison, the stable-event identity tax, and its quantile-crossing form.

Let P and Q be finite output distributions associated with two actions. A memoryless controlled channel requires that, under every adaptive action process,

$$\Pr(Y_t = y \mid A_{1:t}, Y_{1:t-1}) = P_{A_t}(y), \quad P_P = P, \quad P_Q = Q. \quad (1)$$

We compare two exact latent contracts through horizon n .

In the *synchronous stable-event* contract, one seed fixes a table

$$F = (F_t(a))_{1 \leq t \leq n, a \in \{P, Q\}}, \quad (2)$$

where the coordinate indexed by the same pair (t, a) is literally the same latent event across all possible histories. Exactness is equivalent to the transversal constraints

$$\text{Law}(F_1(a_1), \dots, F_n(a_n)) = \bigotimes_{t=1}^n P_{a_t} \quad (3)$$

for every deterministic action string $a_{1:n}$. Let $C_n^{\text{syn}}(P, Q)$ be the minimum entropy of such a table.

In the weaker *history-indexed causal* contract, the seed fixes responses

$$F_t(h_{t-1}, a), \quad (4)$$

so two unrealized histories may assign different latent events to the same time-action label. Every realized adaptive path must still have exactly the law in Eq. (1). Let $C_n^{\text{causal}}(P, Q)$ denote the minimum seed entropy. We write

$$C_\infty^{\text{syn}} := \lim_{n \rightarrow \infty} \frac{C_n^{\text{syn}}}{n}, \quad C_\infty^{\text{causal}} := \liminf_{n \rightarrow \infty} \frac{C_n^{\text{causal}}}{n}. \quad (5)$$

Subadditivity gives the ordinary synchronous limit; the causal construction below is sufficient for the liminf formulation.

The difference between these contracts is not pathwise correctness. It is the identity relation imposed on events that cannot be jointly observed. We therefore define the *stable-event identity tax*

$$\tau_{\text{id}}(P, Q) := C_\infty^{\text{syn}}(P, Q) - C_\infty^{\text{causal}}(P, Q). \quad (6)$$

The central result is that this tax is controlled by the mismatch of the surprisal spectra after their mean entropy difference has been removed.

For a finite distribution $P = (p_i)$, let

$$U_P = -\log_2 P(X), \quad X \sim P, \quad \mu_P := \text{Law}(U_P). \quad (7)$$

Thus $\mathbb{E}U_P = H(P)$. Write W_1 for the one-dimensional Wasserstein distance.

Synchronous converse.— Fix a positive-probability atom $f = (x_1, z_1, \dots, x_n, z_n)$ of a synchronous table, with atom probability $\Gamma(f)$. Because the same atom belongs to every transversal in Eq. (3), for every subset $S \subseteq [n]$,

$$\Gamma(f) \leq \prod_{t \in S} P(x_t) \prod_{t \notin S} Q(z_t). \quad (8)$$

Minimizing over S separates coordinatewise,

$$\Gamma(f) \leq \prod_{t=1}^n \min\{P(x_t), Q(z_t)\}. \quad (9)$$

Hence

$$-\log_2 \Gamma(f) \geq \sum_{t=1}^n \max\{U_P(x_t), U_Q(z_t)\}. \quad (10)$$

Averaging over the table and minimizing the expected absolute surprisal difference over all couplings gives

$$\boxed{C_\infty^{\text{syn}}(P, Q) \geq \frac{1}{2} [H(P) + H(Q) + W_1(\mu_P, \mu_Q)]}. \quad (11)$$

This is the two-action information-spectrum converse. It is a direct consequence of the fact that one full-table atom must simultaneously fit inside every admissible transversal; related minimum-entropy coupling converses are discussed in Refs. [1, 2].

Causal achievability.— Set $h_\star = \max\{H(P), H(Q)\}$ and $h_{\min} = \min\{H(P), H(Q)\}$. The lower bound

$$C_n^{\text{causal}}(P, Q) \geq nh_\star \quad (12)$$

is immediate: the policy that always selects the higher-entropy action produces a trajectory of entropy $nh_\star$, and that trajectory is a deterministic projection of the complete seed.

The upper bound uses correlations that are placed beyond the causal horizon. For block length L , minimum-entropy coupling estimates imply the existence of a coupling

$$J_L \in \Pi(P^{\otimes L}, Q^{\otimes L}) \quad (13)$$

with

$$H(J_L) \leq Lh_\star + A\sqrt{L} + c_0, \quad (14)$$

where $A = \frac{1}{2}(\sqrt{V_P} + \sqrt{V_Q})$, $V_P = \text{Var}(U_P)$, and c_0 is a universal additive constant obtainable from profile/greedy coupling bounds [3].

Let $n = mL$. Divide a P stream $X_{1:n}$ and an auxiliary Q stream $W_{1:n}$ into m blocks. Couple each pair

$$(W^{(j)}, X^{(j+1)}) \sim J_L, \quad j = 1, \dots, m-1, \quad (15)$$

independently, leave the first P block and last Q block independent, and reverse the Q stream before readout. On the r th occurrence of action P output X_r ; on the s th occurrence of action Q output the s th symbol of the reversed W stream.

The key observation is combinatorial. To expose a coordinate of $X^{(j+1)}$, a history needs at least $jL + 1$ occurrences of P . To expose a coordinate of its correlated partner $W^{(j)}$ through the reversed Q stream, it needs at least $n - jL + 1$ occurrences of Q . Seeing both sides would therefore require at least

$$(jL + 1) + (n - jL + 1) = n + 2 \quad (16)$$

actions. No horizon- n history can do so. The dependence is present in the seed but invisible on every admissible causal path.

Consequently every deterministic action word, and hence every deterministic adaptive policy, sees exactly the product law required by Eq. (1). The seed entropy satisfies

$$H(S_n) \leq nh_\star + Lh_{\min} + (m-1)(A\sqrt{L} + c_0). \quad (17)$$

Taking $m, L \rightarrow \infty$ gives

$$\boxed{C_\infty^{\text{causal}}(P, Q) = h_\star}. \quad (18)$$

For example, choosing $L \asymp n^{2/3}$ gives the finite-horizon residual $C_n^{\text{causal}} - nh_\star = O(n^{2/3})$.

Combining Eqs. (11) and (18) produces the promised identity-tax bound,

$$\boxed{\tau_{\text{id}}(P, Q) \geq \frac{1}{2} [W_1(\mu_P, \mu_Q) - |H(P) - H(Q)|]}. \quad (19)$$

Because the means of μ_P and μ_Q are $H(P)$ and $H(Q)$,

$$W_1(\mu_P, \mu_Q) \geq |H(P) - H(Q)|, \quad (20)$$

so the right-hand side is nonnegative. The formula separates a common absolute entropy level from the remaining relative mismatch of the surprisal spectra. Figure 1 summarizes the two mechanisms: the tax is the crossing part of the surprisal geometry, while causal achievability stores correlations beyond the horizon- n anti-diagonal.

In the equal-entropy case,

$$\boxed{H(P) = H(Q) = h \implies \tau_{\text{id}}(P, Q) \geq \frac{1}{2} W_1(\mu_P, \mu_Q)}. \quad (21)$$

Moreover $W_1(\mu_P, \mu_Q) = 0$ precisely when the nonzero probability spectra agree up to permutation. Thus equal Shannon entropy does not remove the tax unless the atom spectra themselves coincide.

A concrete crash test makes the separation visible. Let

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right), \quad Q = (q, q, 1-2q), \quad (22)$$

where $q \in (1/3, 1/2)$ is chosen so that $H(Q) = 3/2$ bits. Numerically,

$$q = 0.4100324281819856\dots, \quad H(P) = H(Q) = 1.5, \quad (23)$$

and

$$W_1(\mu_P, \mu_Q) = 0.4569309374910615\dots \quad (24)$$

Therefore

$$C_\infty^{\text{causal}} = 1.5, \quad C_\infty^{\text{syn}} \geq 1.728465468745531, \quad (25)$$

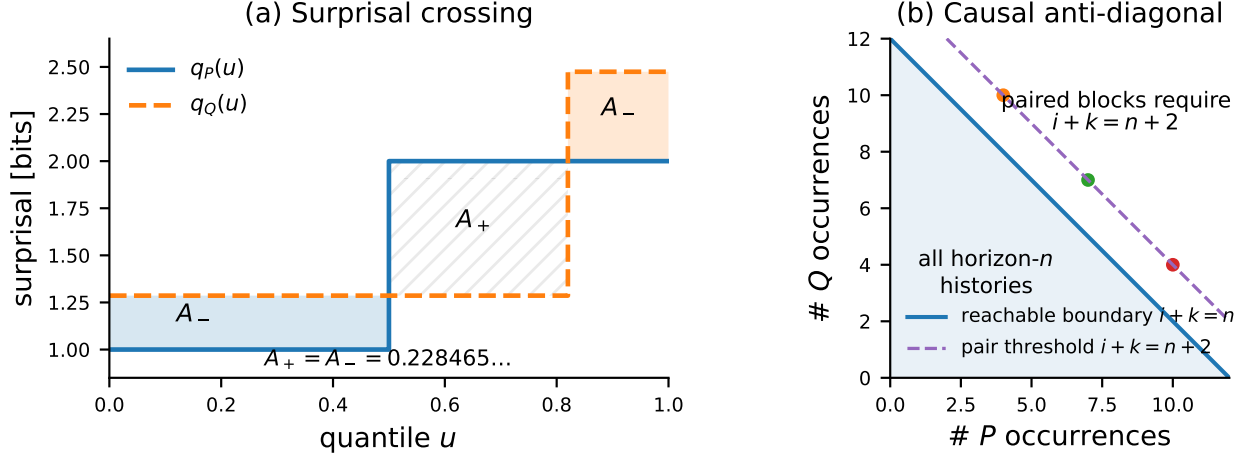


FIG. 1. Two complementary geometries behind the separation. (a) For the equal-entropy crash test, the surprisal quantile curves cross; the two oriented areas are equal and each equals $W_1/2 = 0.228465\dots$ bits. The general lower bound is the smaller oriented area. (b) In the causal construction, a correlated block pair requires at least $i+k = n+2$ action occurrences to expose both components, whereas every horizon- n history satisfies $i+k \leq n$. Correlation is therefore present in the seed but inaccessible on every allowed path.

and hence

$$\tau_{\text{id}} \geq 0.228465468745531 \text{ bits/symbol.} \quad (26)$$

The two latent models reproduce exactly the same family of adaptive observable laws. The positive rate prices only the stronger statement that an unrealized event carrying a given time-action label must retain its identity across incompatible histories.

The limiting equality in Eq. (18) should not be mistaken for finite-horizon attainment. For the same equal-entropy pair, write

$$E_n := C_n^{\text{causal}} - 1.5n. \quad (27)$$

The one-step minimum-entropy coupling already gives $E_1 = 0.5539185856\dots > 0$, and the finite-horizon prefix bound keeps E_n positive, while the anti-diagonal construction yields $E_n = O(n^{2/3})$. Hence

$$0 < E_1 \leq E_n = o(n). \quad (28)$$

The Shannon floor is therefore an asymptotic rate that need not be attained at any finite horizon. This non-attainment is operationally important: the tax concerns an extensive rate under stable cross-history identity, not the disappearance of every finite counterfactual overhead.

For orientation, the crash test obeys the contract hierarchy

$$1.5 = C_\infty^{\text{causal}} < 1.728465\dots \leq C_\infty^{\text{syn}} \leq 1.939147\dots < 2.053918\dots, \quad H(F_{1:2}) = 3.878295576158564\dots, \quad (29)$$

where the final number is the one-step complete-table minimum-entropy coupling. Each inequality prices a

different requirement; none may be moved to another contract without changing the problem.

Equation (19) has a useful quantile-crossing form. Let $q_P(u) = F_{\mu_P}^{-1}(u)$ and $q_Q(u) = F_{\mu_Q}^{-1}(u)$, and define

$$A_+ = \int_0^1 [q_P(u) - q_Q(u)]_+ du, \quad A_- = \int_0^1 [q_Q(u) - q_P(u)]_+ du. \quad (30)$$

Then $W_1 = A_+ + A_-$ and $H(P) - H(Q) = A_+ - A_-$. Therefore

$$\tau_{\text{id}}(P, Q) \geq \min\{A_+, A_-\}. \quad (31)$$

The lower bound is thus exactly the smaller of the two oppositely oriented areas between the surprisal quantile curves. It vanishes precisely when the two curves are ordered almost everywhere; equivalently, one surprisal law first-order stochastically dominates the other. For equal entropies, such ordering forces equality of the surprisal laws, recovering the strictness statement above. This form makes the mechanism geometric: a mere displacement of the mean surprisal is already paid by the Shannon floor; only the *crossing* of the residual spectra is charged as stable-event identity.

The tax is also not an artifact of forcing independent one-time counterfactual tables. Synchronous tables can exploit hidden temporal synergy while preserving every selectable transversal. For the equal-entropy crash test above, a verified sparse two-step table with 23 positive atoms has entropy

$$H(F_{1:2}) = 3.878295576158564\dots, \quad (32)$$

so subadditivity gives

$$C_\infty^{\text{syn}} \leq 1.939147788079282\dots \quad (33)$$

Thus the concrete identity tax is bracketed by

$$\boxed{0.228465468745531 \leq \tau_{\text{id}} \leq 0.439147788079282} \quad (34)$$

for this pair. The same witness has strong dependence between the complete one-time counterfactual tables even though individually selectable future coordinates remain pairwise independent. Correlation can therefore reduce the synchronous bill, but it does not erase the positive rate forced by stable cross-history identity [7].

The converse is also robust to removing many counterfactual transversals. Let $\mathcal{G}_n \subseteq \{P, Q\}^n$ be only the action strings on which a synchronous table is required to be exact, and let $C_n^{\text{syn}}(\mathcal{G}_n)$ denote the corresponding minimum entropy. For a full-table atom define

$$M_t := \max\{U_P(x_t), U_Q(z_t)\}, \quad \Delta_t := |U_P(x_t) - U_Q(z_t)|, \quad (35)$$

and let $s \in \{P, Q\}^n$ choose the more surprising action coordinatewise. The same atom argument gives the pointwise bound

$$-\log_2 \Gamma(f) \geq \sum_{t=1}^n M_t - d_\Delta(s, \mathcal{G}_n), \quad (36)$$

where d_Δ is weighted Hamming distance to the closest required transversal. If the supports of P, Q are positive and

$$D := \max_{x,z} |U_P(x) - U_Q(z)| < \infty, \quad (37)$$

then $d_\Delta \leq Dd_H$. Writing

$$r(\mathcal{G}_n) := \max_s \min_{g \in \mathcal{G}_n} d_H(s, g) \quad (38)$$

for the covering radius, averaging Eq. (36) yields

$$\boxed{C_n^{\text{syn}}(\mathcal{G}_n) \geq nT(P, Q) - Dr(\mathcal{G}_n)}, \quad (39)$$

with

$$T(P, Q) = \frac{1}{2}[H(P) + H(Q) + W_1(\mu_P, \mu_Q)]. \quad (40)$$

Thus $r(\mathcal{G}_n) = o(n)$ preserves the full synchronous information-spectrum rate even if most action strings

are not explicitly required. Counterfactual breadth is therefore a covering geometry, not merely the cardinality of a policy family. This robustness is useful conceptually: the extensive surcharge is tied to how densely stable event identities constrain mutually incompatible futures, not to the literal presence of all 2^n labels.

This separation also clarifies what the theorem does emphnot say. It does not establish that observed randomness is ontic, identify a fundamental substrate, or rule out simulation. Nor does Eq. (18) construct one projectively consistent infinite random tree whose finite restrictions are simultaneously rate-optimal: the anti-diagonal source is horizon dependent. The result is finite-horizon and contract relative. Its content is narrower and operational: exact observable equivalence does not imply equality of realization resources.

The broader lesson is that counterfactual structure has a measurable information geometry. Shannon entropy fixes the cost of realized uncertainty; the Wasserstein mismatch of surprisal spectra measures an additional obstruction when mutually exclusive futures are forced to share stable latent event identities. A causal implementation can hide correlations behind the horizon. A synchronous implementation cannot. The bill is the difference.

-
- [1] M. Kovačević, I. Stanojević, and V. Šenk, On the entropy of couplings, *Inf. Comput.* **242**, 369 (2015).
 - [2] Y. Y. Shkel and A. K. Yadav, Information spectrum converse for minimum entropy couplings and functional representations, *Proc. IEEE Int. Symp. Inf. Theory (ISIT)* (2023).
 - [3] S. Compton, D. Katz, B. Qi, K. Greenewald, and M. Kocaoglu, Minimum-entropy coupling approximation guarantees beyond the majorization barrier, *Proc. AISTATS, PMLR* **206**, 10445 (2023).
 - [4] G. R. Kumar, C. T. Li, and A. El Gamal, Exact common information, *arXiv:1402.0062* (2014).
 - [5] L. Yu and V. Y. F. Tan, Exact channel synthesis, *arXiv:1810.13246* (2018).
 - [6] R. Cont and F. R. Lim, Causal transport on path space, *arXiv:2412.02948* (2024).
 - [7] J. Mikulík, Who Pays the Bill? Substrate Skepticism, Stable Counterfactual Identity, Hidden Synergy, and the Causal Shannon Floor, preprint (2026).